

What Drives Informed Trading Before Public Releases? Evidence from Natural Gas Inventory Announcements

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First version: March 20, 2015

This version: March 25, 2018

Journal of Futures Markets, Forthcoming

Abstract

This paper shows evidence of informed trading in the natural gas futures market before gas inventory announcements. We examine whether traders can predict the upcoming announcement by processing public information. The results show that the difference between the median forecast of analysts with high historical forecasting accuracy and the consensus forecast can be used to predict inventory surprises. This predictor explains some of the pre-announcement price drift, suggesting that informed trading before the announcement is likely to be driven by superior forecasting rather than by information leakage. A simple trading strategy conditioned on the predictor would have generated an annualized Sharpe ratio of 1.26.

JEL classification: D82; D84; G12; G14; G17; Q40

Keywords: Natural gas futures; Inventory announcements; Pre-announcement effect; Informed trading; Inventory forecasting

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We thank seminar participants at West Virginia University and an anonymous referee for helpful comments and suggestions. Errors or omissions are our responsibility.

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1. Introduction

Security prices contain information. Private information is reflected in prices through trading. Public information is generally believed to be incorporated in prices before anyone can trade on it (e.g., French and Roll, 1986). We examine whether some traders are able to make profitable trades by processing public information. Specifically, we analyze futures returns and trading activity around releases of the Weekly Natural Gas Storage Report. This report contains information about U.S. inventory of natural gas. The report is received by all interested parties simultaneously and can be viewed as typical public information. However, we find evidence of informed trading before the release of the report. About 90 minutes prior to the official announcement time, natural gas futures prices begin to drift in the direction of the eventual market reaction to the announcement. The pre-announcement returns and order flow in natural gas futures are significantly correlated with inventory surprises based on the consensus analyst forecast, suggesting that informed traders correctly anticipate inventory news.

Similar evidence of pre-announcement informed trading has been reported in recent studies of macroeconomic announcements. Bernile, Hu and Tang (2016) show evidence of informed trading during the lockup periods before policy announcements of the Federal Open Market Committee (FOMC). Abnormal order imbalances in the stock index futures market are correlated with the subsequent policy surprise. Kurov, Sancetta, Strasser and Wolfe (2017) find strong evidence of price drift in equity index and Treasury futures prior to the official release of several major U.S. macroeconomic announcements. On average, the pre-announcement price move accounts for almost half of the total price adjustment around the announcement. Similar evidence of price drift before U.K. macroeconomic news has attracted intense media attention.¹

¹ See, for example, Bird (2017) and Bingham (2017).

Our evidence of informed trading before the Weekly Natural Gas Storage Report announcements shows that the pre-announcement effect is not limited to major macroeconomic announcements.

One can think of several channels through which informed traders might obtain informational advantage before the release of the official data. In the first channel, some traders might get advance access to the news due to information leakage. Bernile et al. (2016) show that informed trading before FOMC announcements is associated with the FOMC's embargo practice, suggesting that information about monetary policy decisions may be leaked by parties who have early access to FOMC statements.² Similarly, Kurov et al. (2017) find stronger evidence of informed trading before macroeconomic announcements released under less secure release procedures. This raises the possibility that access to leaked information gives some traders an unfair advantage.

In the second channel, traders may become informed about upcoming announcements by collecting proprietary information. For example, institutional traders may conduct informal surveys of natural gas storage operators and use the collected data to forecast inventory surprises. Finally, traders might be able to predict announcement surprises through the skillful processing of public information. Kim and Verrecchia (1994) contend that some traders can process public information better than others. This increases information asymmetry around public announcements. If some traders are able to use their superior information processing ability to predict inventory surprises and trade on their predictions, we should observe a price drift in the "right" direction before the announcement. Kurov et al. (2017) examine this possibility for macroeconomic announcements but are unable to produce clear evidence that the pre-announcement drift is explained by smart reprocessing of publicly available data.

² In a related paper, Cieslak, Morse and Vissing-Jorgensen (2018) argue that the Fed tends to release information about future policy decisions through informal communication with the media and the financial sector.

We offer evidence on the public information channel of the pre-announcement informed trading by looking at whether such trading is related to inventory forecasts of analysts with superior historical forecasting ability. Bloomberg assigns ranks to three best analysts based on their past forecasting accuracy. We use the median of the ranked analyst forecasts as a proxy for the “informed” forecast. We then compute a simple predictor of the inventory surprise as the difference between the “informed” forecast and the regular Bloomberg consensus forecast. Individual analyst forecasts and ranks are available to Bloomberg subscribers before the announcement. Thus, our predictor is based on what effectively is public information.³ Ben-Rephael, Da and Israelsen (2017) note that Bloomberg subscribers are much more likely to be institutional investors than retail traders. Hendershott, Livdan and Schürhoff (2015) provide evidence that institutional trading predicts news announcements, suggesting that institutions are informed traders. They have expertise and resources to process available information and optimally trade on it.

We show that the predictor discussed above has explanatory power for the natural gas inventory surprises and for the pre-announcement futures returns, both in- and out-of-sample. This finding is consistent with pre-announcement informed trading based on superior forecasts. For example, suppose that the consensus forecast predicts inventory increase of one billion cubic feet, while analysts with superior historical forecasting accuracy forecast an increase of two billion cubic feet. The difference between these two forecasts can be viewed as a forecast of what traders relying on the Bloomberg consensus will view as the inventory surprise. If some traders sell futures contracts before the announcement based on this forecast of the inventory shock, prices will decline, leading to a pre-announcement drift. Therefore, we provide evidence

³ The number of Bloomberg subscribers is about 320,000 globally. It is reasonable to assume that most large futures traders have access to Bloomberg.

that some traders use what effectively is public information to make well-timed trades before public announcements.

It is possible that the pre-announcement informed trading is driven in part by information leakage or by proprietary data collection. However, our results show that informed trading due to superior processing of public information is a plausible explanation for the pre-announcement price drift. We propose a simple trading strategy conditioned on the predictor to capture the pre-announcement drift. The trading strategy consists of buying (selling) natural gas futures contracts 90 minutes before the release time if the predictor is negative (positive), and closing the position before the announcement. This strategy produces an annualized Sharpe ratio of 0.84 in the full sample of announcements and an annualized Sharpe ratio of 1.26 on days with large values of the predictor.

2. Data

2.1. The Weekly Natural Gas Storage Report

The U.S. Energy Information Administration (EIA) releases its Weekly Natural Gas Storage Report every Thursday at 10:30 a.m. ET. In some weeks, which include holidays, releases are delayed by one day. These reports include estimates of the volume of working natural gas held in underground storage facilities in the lower 48 states and the net change in storage for the week ending the previous Friday. These estimates of working gas are derived from a survey of a sample of operators of underground storage facilities. Approximately 140 operators report for their operations at about 400 storage fields. Respondents submit estimates of working gas in storage fields operated by their company as of 9 a.m. on Friday each week by 5 p.m. Eastern Time the following Monday. Several studies show that unexpected changes in gas inventory

have a significant impact on natural gas futures prices immediately after the announcement.⁴ In contrast, our study focuses on the natural gas futures prices and trading activity before the official announcement time.

2.2. Sample Selection

Our sample period is from July 2008 through January 2016. July 2008 serves as the beginning point because in mid-2008 the EIA modified its weekly natural gas storage survey to reflect changes in the industry and improve the accuracy of storage estimates (Halova et al., 2014). The improvement of the information collection process decreased the noise in the natural gas storage estimates. Our sample period includes 387 releases of the Weekly Natural Gas Storage Report. To minimize the potentially confounding effects of the EIA's Weekly Petroleum Status Report on natural gas futures, we exclude 53 occasions when the two reports were released on the same day.⁵ The final sample contains 334 observations.

2.3. Inventory Surprises

Following previous studies, we use the Bloomberg consensus forecast to measure the market expectation of the changes in inventory. Every week, around 20 analysts submit their forecasts of the change in natural gas inventory.⁶ Bloomberg then computes the consensus forecast as the median of the analyst forecasts.⁷ Bloomberg normally ranks the top three analysts based on their forecasting accuracy in the previous 24 months.⁸ Table I lists all active analyst firms during the

⁴ See, for example, Linn and Zhu (2004), Gay, Simkins and Turac (2009), and Halova, Kurov and Kucher (2014).

⁵ The Weekly Petroleum Status Report that includes inventory estimates for crude oil and oil products is released every Wednesday at 10:30 a.m. ET. In some weeks, which include holidays, releases are delayed by one day.

⁶ The average time between the date of forecast submission and the release date is 1.15 days. If forecasts submitted more than 2 days before the release date are dropped from calculation of the consensus forecast, the empirical results are similar. These results are available upon request.

⁷ The empirical results based on the arithmetic average of analyst forecasts, available upon request, are similar.

⁸ Bloomberg's analyst ranking methodology is described in section 5.1.

sample period.⁹ As shown in the table, there are substantial cross-sectional differences in forecasting ability measured as the mean standardized absolute forecast error.¹⁰ In addition, the group of the ranked analysts is quite stable. Only five analysts appeared in the group of the top three forecasters after more than 50 inventory announcements.

[Insert Table I here]

We use the median value of the ranked analyst forecasts to construct the “informed” forecast. The ranking information is updated immediately after the release. To avoid forward-looking bias, we use the analyst ranks from the previous week. Bloomberg subscribers have access to this information prior to the release of the Weekly Natural Gas Storage Report.

The inventory surprise, S_t , is computed as the difference between the actual change in inventory and the consensus forecast:

$$S_t = A_t - F_t, \quad (1)$$

where A_t is the actual inventory change and F_t is the consensus forecast or the “informed” forecast for week t .¹¹

Summary statistics for the inventory surprises are shown in Table II. The table shows that the standard deviation of the inventory surprises based on ranked analyst forecasts is lower than that of the surprises based on the consensus forecast. Moreover, the inventory surprises based on the “informed” forecast tend to be smaller in absolute value. These statistics indicate that the “informed” forecast tends to be more accurate than the consensus forecast.

⁹ The table only includes 41 analyst firms that submitted more than 50 forecasts during our sample period. There were 85 different forecast submitters during this period.

¹⁰ The standardized absolute forecast error for analyst i in week t is defined as $|A_t - F_{i,t}|/\sigma_t$, where A_t is the actual announcement, $F_{i,t}$ is the individual forecast and σ_t is the standard deviation of forecasts for week t .

¹¹ Chang, Daouk and Wang (2009) use a similar approach to measure inventory surprises for crude oil. We also computed inventory surprises scaled by the storage level to control for the potential effects of trends in inventory. The empirical results, available upon request, are similar.

[Insert Table II here]

Figure 1 shows natural gas inventory and inventory surprises over the sample period. As seen in Panel A of Figure 1, the storage level shows a strong seasonality. It tends to increase from April to October (injection season) and decrease from November to March (withdrawal season). However, Panel B shows that the inventory surprises, which represent unexpected changes in the natural gas inventory, do not have a clear seasonal pattern.

[Insert Figure 1 here]

3. Evidence of Pre-announcement Informed Trading

To look for evidence of pre-announcement informed trading, we first graph the cumulative average returns (CARs) of the nearby natural gas futures in an intraday event window surrounding the announcement.¹² Positive surprises represent bad news for natural gas futures prices and negative surprises represent good news. Following Bernile et al. (2016), we invert the sign of returns for bad news. This allows aggregating CARs for positive and negative surprises. Panel A of Figure 2 plots the CAR in the interval from midnight of the announcement day to three hours after the announcement. The figure shows that about 90 minutes before the announcement natural gas futures prices begin moving gradually in the direction of the subsequent post-announcement returns.¹³ As evidenced by the 90% confidence interval, the pre-announcement returns are statistically significant.

[Insert Figure 2 here]

¹² Since the nearby contract becomes relatively illiquid shortly before expiration, we switch to the next contract in the last three days of trading of the nearby contract.

¹³ The correlation of returns in the $[t - 90min, t - 5min]$ and $[t - 5min, t + 30min]$ windows, with time t being the announcement time, is positive and statistically significant at the 5% level.

We also use second-by-second futures data to examine cumulative order imbalances. If pre-announcement informed trading exists, we should find order imbalances in the “right” direction before the release time. We use the tick rule to classify trading volume as buyer- or seller-initiated.¹⁴ Similar to returns, order imbalances for positive inventory surprises are multiplied by minus one. Panel B of Figure 2 plots the average cumulative order imbalances from 90 minutes before to 30 minutes after the release. The figure shows that the pre-announcement order imbalances are in the direction of the subsequent post-announcement trading. The cumulative order imbalance before the release accounts for about one-third of the total order imbalance around the announcement on average.

Informed traders may choose to trade on their forecasts of the announcement surprise days before the announcement. Figure 3 plots daily cumulative average returns in the natural gas futures conditional on the inventory surprise from four days before to the day after the announcement. The CAR is not statistically significant until the announcement day. This suggests that the informed trading is concentrated immediately before the announcement.

[Insert Figure 3 here]

The Natural Gas Storage Report is released at 10:30 a.m. ET. As shown in Figure 2, the pre-announcement drift begins about 90 minutes before the announcement, which coincides with the beginning of open outcry trading in the natural gas futures market. Although the large majority of futures trading during our sample period occurred on the 24-hour electronic trading platform called Globex, trading activity increases dramatically after 9:00 a.m. Figure 4 plots the average trading volume around the official release time. As seen in the figure, the volume rises significantly 90 minutes before the announcement. The high level of trading activity on Globex

¹⁴ Specifically, volume in a 1-second bar is classified as buyer-initiated if the price for that bar is higher than the last different price. Otherwise, the volume in the bar is classified as seller-initiated.

after 9:00 a.m. allows informed traders to hide their trading. Therefore, even if an informed trader receives tradable information several hours before the release of the Natural Gas Storage Report, it may be optimal to trade on it only shortly before the announcement. This is consistent with Collin-Dufresne and Fos (2015, 2016) who show that informed traders are more active when uninformed trading is high. It is also possible that investors optimally choose what signals to pay attention to, and the 90 minutes prior to the announcement may be a time when natural gas futures traders process storage information that might have been available before (Admati and Pfleiderer, 1988; Kacperczyk, Van Nieuwerburgh, and Veldkamp, 2016). Minimizing risk is another possible reason for delaying trading until shortly before the announcement. Establishing a position long before the announcement would expose the trader to potential losses driven by news unrelated to the inventory announcement.

[Insert Figure 4 here]

We examine the explanatory power of inventory surprises for the returns around the announcement by estimating the following regression:

$$R_t = a_0 + a_1 S_t + e_t, \quad (2)$$

where R_t is the continuously compounded natural gas futures return over a given intraday event window, S_t is the inventory surprise, and e_t is an i.i.d. error term.¹⁵

We first estimate this regression using the event window from 5 minutes before to 30 minutes after the announcement. Table III presents the regression results. Consistent with previous literature, the natural gas futures market negatively reacts to inventory shocks. The table also shows that forecasts of more accurate analysts are given more weight by the market.

¹⁵ As a robustness check, we also run this regression using the return of the U.S. Natural Gas ETF (ticker symbol UNG) as the dependent variable. The results, available upon request, are similar to those based on the futures returns.

Specifically, the inventory surprise based on the “informed” consensus forecast explains a larger fraction of variation in the futures returns compared to the surprise based on the Bloomberg consensus. Halova et al. (2014) show that noise in inventory surprises leads to a downward bias in the market response coefficients and lowers the regression R^2 . The larger estimate of the response coefficient and higher R^2 for the regression using inventory surprises based on the “informed” consensus forecast suggest that these forecasts are less noisy than the regular consensus forecasts computed using data for all analysts.

[Insert Table III here]

Since we are interested in the pre-announcement effect, we then estimate the event study regression using futures returns in the interval from 90 minutes before to 5 minutes before the announcement using the inventory surprises based on the Bloomberg consensus.^{16,17} Table IV provides the regression results. Consistent with the existence of pre-announcement informed trading, futures returns before the announcement are significantly and negatively related to the inventory shocks. We also estimate the same regression using the order imbalance in the pre-announcement event window as the dependent variable. Table IV shows that the coefficient estimate on the inventory surprise is negative and statistically significant. This result is consistent with informed trading based on forecasts of inventory surprises.

[Insert Table IV here]

¹⁶ We use 5 minutes before the official release time as the end of the pre-announcement window to minimize the influence of high-frequency traders. Flynn (2012) and Dicolo and Rogow (2012) discuss instances when high-frequency traders send a flood of orders in an effort to trigger large price swings just before the EIA storage data is announced. They then make profits in volatile trading by taking advantage of their superior speed.

¹⁷ The pre-announcement informed trading can also be examined by regressing inventory surprises on pre-announcement returns or order imbalances. However, this specification is interchangeable with equation (2) in terms of statistical inference. Following Bernile et al. (2016) and Kurov et al. (2017), we choose to regress the returns and order imbalances on inventory surprises.

To investigate the sensitivity of the regression results to the length of the event window, we re-estimate the regression in equation (2) for different event windows. We move the starting point of the window used to compute the futures returns from 150 minutes before the announcement to 10 minutes before the announcement at five-minute increments. The endpoint of the event window is fixed at five minutes before the announcement. As shown in Figure 5, the estimated coefficient of the inventory surprise based on the Bloomberg consensus declines for the pre-announcement windows that start less than 90 minutes before the release time. This is consistent with our finding in Figure 2 showing that the pre-announcement CAR becomes significant about 90 minutes ahead of the announcement.

[Insert Figure 5 here]

The intercept coefficient estimates in Tables III and IV are negative and statistically significant. This suggests that natural gas futures prices declined in our sample period. This result is mainly driven by a large decline in natural gas prices during first year of our sample.¹⁸ If we exclude the period from July 2008 to July 2009, the intercept estimates become much smaller but the slope coefficient estimates remain similar to those in Tables III and IV.¹⁹

4. Sources of Informed Trading

4.1 Informed Trading Based on Superior Forecasts

A possible explanation for the pre-announcement effect documented in the previous section is that some traders are able to predict inventory surprises by processing available information and

¹⁸ The price of natural gas declined by approximately 70% between July 2008 and July 2009 and remained low for the rest of our sample period.

¹⁹ These results are not tabulated for brevity. In an additional robustness check, we estimate a two-year rolling regression of pre-announcement returns on inventory surprises. The results (not reported but available upon request) show that the relation between the inventory surprises and pre-announcement returns is negative and significant for all two-year sub-periods in our sample.

trade on their predictions before the announcement. We have shown that forecasts of analysts with best prior forecasting accuracy are superior to the consensus forecast. If the Bloomberg consensus forecast reflects the broad market expectations of natural gas inventory, it is reasonable to use the median value of the ranked analyst forecasts as a proxy for expectations of informed traders. Therefore, the difference between the “informed” and the simple consensus forecasts can be used as a predictor of the inventory surprise:

$$P_t = F_t^I - F_t^C, \quad (3)$$

where F_t^I and F_t^C are the “informed” and consensus forecasts of inventory change, respectively. Importantly, this predictor uses only information available to Bloomberg subscribers before the announcement.

To formally explore the predictive power of our predictor of the surprise, we run the following OLS regression:

$$S_t = a_0 + a_1 P_t + \sum_{n=1}^L b_n S_{t-n} + e_t. \quad (4)$$

Lags of inventory surprises are included in the model to control for potential seasonality. The optimal number of lagged inventory surprises based on the Akaike information criterion is two. In addition to $L=2$, we also use model specifications with one lag, as well as a model with no lags of inventory surprises. As shown in Table V, the coefficient estimate of the predictor is statistically significant and close to one for all of these models. This indicates that there is a strong relation between the predictor and the inventory surprise.

[Insert Table V here]

Next, we examine whether the pre-announcement return is related to the predictor. Panel A of Table VI shows that the predictor has significant explanatory power for the pre-announcement returns. The negative sign of the coefficient of the predictor is consistent with

informed trading based on superior forecasts before the announcement. When informed traders believe that the consensus forecast overestimates the increase or underestimates the decrease in inventory, they buy futures contracts before the release time, leading to a positive price drift. When the consensus forecast underestimates the increase or overestimates the decrease in inventory, informed traders sell futures contracts before the announcement, leading to a negative price drift. Panel B reports estimates of a similar regression with the pre-announcement order imbalance used as the dependent variable. These estimates are qualitatively similar to those in Panel A of the table.

[Insert Table VI here]

A natural follow-up question is whether the information contained in the predictor is completely incorporated into the market price by the time of the announcement. According to the efficient market hypothesis, since information contained in the predictor is widely available before the announcement, the predictor should be unrelated to price changes after the release of the inventory report. Regression results reported in Table VI show that the return and order imbalance after the announcement are not significantly related to the predictor. This finding suggests that the information contained in the superior forecast predictor is fully reflected in futures prices before the announcement. This may be due to competition of multiple traders who trade on similar information (Holden and Subrahmanyam, 1992).

Pre-announcement informed trading may be related to other types of public information. For example, Mu (2007) shows that temperature shocks have a significant impact on returns and volatility of natural gas futures. Traders may use temperature shocks to forecast natural gas inventory changes. Traders may also be able to use information about crude oil inventory, which is released one day before the gas storage report, to predict natural gas inventory surprises. In

untabulated results, we show that temperature shocks cannot be used to predict natural gas inventory surprises.²⁰ Similarly, we find no evidence that crude oil inventory surprises can be used to predict subsequent natural gas inventory surprises.

4.2. Anchoring Bias in Inventory Forecasts

Campbell and Sharpe (2009) show that consensus forecasts of macroeconomic announcements are anchored to the recent past, which results in the announcement surprises being predictable. The anchoring bias can be illustrated by the following equation:

$$F_t = \lambda E[A_t] + (1 - \lambda)\bar{A}_h, \quad (5)$$

where $E[A_t]$ is the unbiased forecast and $\bar{A}_h$ is the average of the last h realizations. If $\lambda < 1$, the forecast is affected by anchoring bias. Substituting $E[A_t] = E[S_t] + F_t$ (obtained from equation (1)) produces a simple regression test for anchoring bias:

$$S_t = \gamma(F_t - \bar{A}_h) + \varepsilon_t, \quad (6)$$

where $\gamma \equiv (1 - \lambda)/\lambda$. When the estimate of γ is positive and significant (implying $\lambda < 1$), the consensus forecast is systematically biased to historical announcements.

Following Campbell and Sharpe (2009), we use the last actual inventory change ($h = 1$) and the mean of the last three realizations ($h = 3$) to test for anchoring bias in the consensus forecast. The results are reported in Panel A of Table VII. The estimate of γ is positive and statistically significant, suggesting that the consensus forecasts contain anchoring bias. Because both the consensus forecast and historical actual values of inventory changes are publicly available before the announcement, this result indicates that traders can predict inventory

²⁰ This analysis uses temperature data for Texas, Illinois, Ohio, Michigan, California and New York, which together account for about half of the residential and commercial natural gas consumption in the U.S.

surprises by using public information. Regression results in Panel B of Table VII show that the “informed” consensus is not affected by anchoring bias.

[Insert Table VII here]

Equation (6) shows that inventory surprises include a predictable component $\gamma(F_t - \bar{A}_h)$ and an unexpected component ε_t . Zhou (2015) argues that the predictable component may explain the price drift before macroeconomic announcements. By regressing pre-announcement returns on the two components of the surprise, he finds that the coefficient on the anticipated component is statistically significant. Using a similar regression, we find a similar result. Table VIII shows that the coefficient estimate of the expected component of the inventory surprise is statistically significant. However, when we keep only the expected component of the surprise in the regression, the R^2 becomes trivial. This finding suggests that most of the information influencing price movement during the pre-announcement window is unrelated to anchoring bias in consensus forecasts. This inference is consistent with the conclusion of Hess and Orbe (2013) that the significant relation between announcement surprises and past announcements is a statistical artifact rather than result of a cognitive bias of analysts. Therefore, the anchoring bias predictor does not contain tradable information.

[Insert Table VIII here]

To further alleviate possible concerns that our predictor based on the “informed” consensus and the anchoring bias predictor have similar information content, we use the two predictors simultaneously to forecast the pre-announcement returns and inventory surprises based on the Bloomberg consensus. The results are reported in Table IX. The predictive power of the anchoring bias predictor is dominated by that of our predictor. The estimated coefficient of the anchoring bias predictor is insignificant, while our predictor remains significantly related to

the pre-announcement returns and inventory surprises. These results show that, although sophisticated traders can use anchoring bias in consensus forecasts to predict storage surprises, this cannot explain the pre-announcement price drift. In addition, we show that the forecasting power of our predictor based on the ranked analyst forecasts is not explained by anchoring bias.

[Insert Table IX here]

5. Robustness and Economic Significance

5.1. In-sample Robustness Checks

In the previous sub-sections, we use economist ranks from Bloomberg to construct a predictor of inventory surprises. Bloomberg normally ranks top three analysts and uses a 24-month evaluation period. To test the robustness of our empirical results to the changes in the group of the ranked analysts, we replicate Bloomberg's methodology for computing economist ranks. It allows us to test whether our results are robust to changes in the length of evaluation period and the number of forecasts used to compute the "informed" consensus.

To evaluate an economist's forecasting performance for a given period, Bloomberg standardizes each individual forecast error as follows:

$$Z_{i,t} = \frac{F_{i,t} - A_t}{\sigma_t}, \quad (7)$$

where $F_{i,t}$ is the forecast of economist i on announcement day t , A_t is the announced number, and σ_t is the standard deviation of all economist forecasts. The distribution of standardized forecast errors is assumed to be normal with a mean of $Z_{A,t}$ and standard deviation of one.

$$Z_{A,t} = \frac{\mu_{F,t} - A_t}{\sigma_t}, \quad (8)$$

where $\mu_{F,t}$ is the mean of all economist forecasts for the given period.

The score of a given economist for announcement day t is computed as the probability that the absolute value of a randomly drawn number from the assumed distribution is greater than or equal to the given economist's absolute forecast error $|Z_{i,t}|$:

$$P(|\text{Normal}(Z_{A,t}, 1)| > |Z_{i,t}|) = 1 - \Phi(-Z_{A,t} + |Z_{i,t}|) + \Phi(-Z_{A,t} - |Z_{i,t}|), \quad (9)$$

where $\Phi(\cdot)$ is the cumulative standard normal probability function.

Based on this methodology, if an individual forecast error $(F_{i,t} - A_t)$ is zero, the corresponding score is 100%. Conversely, if the forecast error is very high, the score will be low. Bloomberg ranks economists based on their cumulative scores in the past 24 months.

Table X reports the forecasting power of the predictor for inventory surprises and pre-announcement returns based on shorter evaluation periods (6, 12 and 18 months, rather than 24 months used by Bloomberg) to rank the top three economists.²¹ The empirical results are similar to those based on the ranks provided by Bloomberg.

[Insert Table X here]

We then compute the predictor based on forecasts of the top five and top 10 economists. The regression results based on this modified predictor are shown in Table XI. Consistent with the results in Tables V and VI, the modified predictor has significant predictive power for inventory surprises and pre-announcement returns.

[Insert Table XI here]

In additional robustness checks, we examine whether the predictor of inventory surprises performs differently in injection and withdrawal seasons to address the concern about the impact of the seasonality of consumption of natural gas. We also use the weighted least squares method

²¹ As before, to avoid forward-looking bias, we use the economist ranks from the previous week. For economists who did not provide a forecast in the previous week, we use the most recent ranks before the previous week.

of Yohai (1987) to estimate our regressions. This so-called MM estimator is robust in the presence of outliers. To alleviate the potential concerns about small-sample issues, we use a simple bootstrap exercise following Lucca and Moench (2015). The results of these robustness checks, available upon request, are similar to the reported results.

5.2. Out-of-sample Tests

Welch and Goyal (2008) find that return predictive regressions with good in-sample performance tend to provide little forecasting power out-of-sample. Specifically, popular equity premium predictors cannot beat an unconditional historical average equity premium. We perform an out-of-sample test to examine whether our superior forecast predictor provides incremental predictive power for the pre-announcement returns and storage surprises relative to the prevailing mean benchmarks.

Following Rapach, Strauss and Zhou (2010), we use an expanding estimation window to calculate out-of-sample forecasts of the inventory surprise and of the pre-announcement return:

$$\hat{Y}_{m+1} = \hat{a}_{0,m} + \hat{a}_{1,m}P_{m+1}, \quad (10)$$

where m is the number of observations in the estimation window, $\hat{a}_{0,m}$ and $\hat{a}_{1,m}$ are the coefficient estimates generated by regressing $\{Y_t\}_{t=1}^m$ on a constant and $\{P_t\}_{t=1}^m$. Y_t is either the natural gas inventory surprise or the pre-announcement return. To make sure that the results are robust to the choice of the estimation window, we alternatively pick 2010, 2012, and 2014 as the starting point of the out-of-sample period.

We then compute the out-of-sample R^2 statistic, R_{OS}^2 , as the proportional reduction in the mean square forecast error (MSFE) for the predictive regression based on our predictor vis-à-vis the mean benchmark forecast:

$$R_{OS}^2 = 1 - \frac{\sum_{k=q_0+1}^q (Y_{m+k} - \hat{Y}_{m+k})^2}{\sum_{k=q_0+1}^q (Y_{m+k} - \bar{Y}_{m+k})^2}, \quad (11)$$

where $\bar{Y}_{m+k}$ is the historical average of the pre-announcement return or of the inventory surprise before period $m + k$. q_0 and q are the endpoints of the in-sample period and the full sample period, respectively. A positive R_{OS}^2 indicates that our predictor outperforms the prevailing mean.

We use the Clark and West (2007) out-of-sample MSPE-adjusted statistic to test the null hypothesis that $R_{OS}^2 \leq 0$ against the alternative hypothesis that $R_{OS}^2 > 0$. Panel A of Table XII shows the results. The positive and significant R_{OS}^2 statistics suggest that our predictor outperforms the prevailing mean benchmarks for both the inventory surprise and the pre-announcement return.

We further analyze out-of-sample predictability of inventory surprises and pre-announcement returns using a variety of autoregressive models as baseline forecasting models. For consistency, we use the same number of lags as in section 4.1. For each AR model, the competing forecasting model includes our predictor as an additional regressor. Panel B of Table XII shows that models with the predictor consistently outperform baseline models out-of-sample. Therefore, our predictor has significant predictive power both in- and out-of-sample.

[Insert Table XII here]

5.3. Economic Significance

In section 4.1, we show that the price drift before the announcement is related to the predicted value of the inventory surprise. Therefore, it may be worthwhile to use this predictor as a trading indicator during the pre-announcement period. Among the 334 announcements in the sample, there are 63 events with a zero predictor. Therefore, we have 271 announcements with tradable values of the predictor. Panel A of Table XIII shows that a simple trading strategy that consists

of buying (selling) a futures contract 90 minutes before the release time if the predictor is negative (positive), and closing the position 5 minutes ahead of the announcement would have earned an annualized Sharpe ratio of 0.84.²² Investors are primarily concerned about downside risk. Therefore, we also compute the Sortino ratio as follows:

$$\text{Sortino Ratio} = \frac{\bar{R} - T}{\sqrt{\frac{1}{N} \sum_{i=1}^N (\min(0, R_i - T))^2}}, \quad (12)$$

where $\bar{R}$ is the average per-event return, T is the target rate of return that we assume to be equal to zero, R_i is the return for event i , and N is the total number of events in the sample. Our trading strategy would have yielded an annualized Sortino ratio of 1.40.²³

[Insert Table XIII here]

Small values of the predictor may not provide informed traders with profitable trading opportunities. Using the sample of announcements with non-zero values of the predictor, we classify values of the predictor in the first and fourth quartiles as large and the remaining values as small. Panel B in Table XIII reports that the simple trading strategy discussed above would have yielded an annualized Sharpe ratio of 1.26 and an annualized Sortino ratio of 2.12 in the large predictor subsample. Figure 6 depicts the 24-month moving average of the pre-announcement trading returns. This moving average is positive for the entire sample period, suggesting that the positive average returns are persistent.

[Insert Figure 6 here]

²² We compute the annualized Sharpe ratio as the per-event Sharpe ratio multiplied by the square root of the average number of announcements per year.

²³ The Sortino ratio is annualized the same way as the Sharpe ratio.

6. Conclusion

This paper documents informed trading in the natural gas futures market before the Weekly Natural Gas Storage Report announcements. We find significant price changes and order imbalances in the “right” direction relative to the subsequent inventory news. The results show that the pre-announcement informed trading can be explained, at least in part, by processing of easily available information. We construct a predictor of the inventory surprise by using the difference between the median forecast of ranked analysts and the overall consensus of analysts in the Bloomberg survey. We propose a simple trading strategy conditioned on this predictor. The strategy involves buying (selling) futures contracts 90 minutes ahead of the announcement when the predictor is negative (positive), and closing the position right before the announcement. This strategy would have yielded a large Sharpe ratio. Traders may use public information from a variety of sources and employ sophisticated forecasting models or forecast combinations. Such market participants may be able to predict announcement surprises more accurately, which should translate into even better trading performance.

Our results have two important implications. First, our predictor is based on information that is common knowledge before the announcement, suggesting that careful analysis of public information can lead to informational advantage. This finding may alleviate concerns of traders and regulators that pre-announcement informed trading is necessarily driven by information leakage. Second, we relate the pre-announcement returns to superior forecasting, suggesting that trading based on such superior forecasts facilitates price discovery in financial markets and improves market efficiency. Although we look only at the natural gas futures market around the weekly releases of the Natural Gas Storage Report, our findings should be of interest to academics interested in the sources of informed trading in general.

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Table I
Analyst Firms Providing Natural Gas Storage Forecasts

Analyst Firm	Time Period		Obs.		Mean Standardized Absolute Forecast Error
	From	To	Total Obs.	Ranked Obs.	
Societe Generale	01/08/09	10/10/13	205	13	0.94
Strategic Energy & Economic Research	08/12/10	01/04/13	92	12	0.94
JP Morgan Chase Bank	07/03/08	01/14/16	333	323	0.94
Macquarie Energy	02/17/11	01/14/16	216	78	0.96
SunTrust Robinson Humphrey	07/17/08	09/02/10	95	23	0.96
Noble Americas	03/31/11	08/08/13	94	39	0.98
IAF Advisors	07/03/08	01/14/16	299	119	0.99
Louis Capital Markets	02/05/09	01/06/11	77	0	1.01
Conoco Phillips	07/03/08	01/14/16	310	157	1.03
Macquarie Bank	04/09/09	02/10/11	82	0	1.03
RBC Capital Markets	07/03/08	03/25/10	63	8	1.04
Citigroup Global Markets	02/16/12	01/14/16	155	49	1.05
DTE Energy Trading Inc	09/13/12	10/08/15	129	7	1.10
ICAP Energy	07/10/08	12/31/15	282	5	1.12
SG Americas Securities	05/22/14	01/07/16	61	0	1.13
Wells Fargo Securities	07/03/08	01/07/16	304	76	1.14
Credit Suisse	02/17/11	03/20/14	126	0	1.15
Energy Aspects Ltd	09/11/14	01/14/16	53	0	1.16
Mitsui & Co Commodity Risk	06/06/13	09/03/15	92	0	1.17
Raymond James & Associates	07/03/08	08/11/11	115	3	1.18
Tradition Financial	03/24/11	04/24/14	57	0	1.20
Cormark Securities	09/17/09	01/14/16	255	0	1.22
Johnson Rice & Co	01/10/13	01/14/16	127	0	1.22
Morgan Keegan & Co. Inc.	07/03/08	05/05/11	110	0	1.25
Asset Risk Management	05/02/13	01/14/16	120	0	1.29
Ritterbusch & Assoc.	07/03/08	03/26/15	279	0	1.31
UBS	07/03/08	06/04/15	280	10	1.32
FirstEnergy Capital	07/03/08	01/14/16	309	0	1.33
Hencorp	07/03/08	04/09/15	269	0	1.37
Fellon-McCord	09/23/10	04/02/15	127	0	1.39
Summit Energy	07/03/08	05/09/13	198	0	1.44
Prestige Economics	09/03/09	12/19/13	184	0	1.44
MF Global	07/03/08	03/15/12	98	0	1.47
Ecova Inc	07/06/12	01/14/16	134	0	1.50
Tuohy Brothers	02/04/10	01/14/16	236	0	1.56
Gelber & Associates	07/03/08	11/13/15	212	0	1.56
Energy Overview	01/27/11	04/30/15	148	0	1.66
Citi Futures Perspective	07/03/08	01/14/16	313	0	1.74
Wavelength CTA	07/24/08	03/01/12	71	0	1.76
C.H. Guernsey	07/03/08	12/31/15	321	0	1.91

The table lists all analyst firms that provided storage forecasts to Bloomberg more than 50 times during the period from July 2008 through January 2016. The firms are sorted by their mean standardized absolute forecast error. Ranked analysts are those ranked by Bloomberg among the top three based on their forecasting accuracy.

Table II
Summary Statistics for Inventory Surprises

	Mean	Median	1 st Quartile	3 rd Quartile	St. deviation
S_t^C	0.42	0.00	-5.00	5.00	7.38
S_t^I	0.65	1.00	-4.00	5.00	6.51
$ S_t^C $	5.78	5.00	3.00	7.00	4.59
$ S_t^I $	5.20	4.50	2.00	7.00	3.96
$ S_t^I - S_t^C $	-0.58***				

This table presents summary statistics for inventory surprises based on the Bloomberg consensus forecast (S_t^C) and on the “informed” consensus (S_t^I). The “informed” consensus is the median value of ranked analyst forecasts. The sample period is from July 2008 to January 2016 and contains 334 observations. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table III
Regression Results for Returns after Inventory Announcements

	Bloomberg Consensus	“Informed” Consensus
Intercept (a_0)	-0.42 (0.11)***	-0.34 (0.10)***
Slope (a_1)	-0.17 (0.02)***	-0.22 (0.02)***
R^2	0.32	0.44

The table reports estimates for the following model: $R_t = a_0 + a_1 S_t + e_t$, where R_t is the continuously compounded futures return (expressed in percent) in the event window from 5 minutes before to 30 minutes after the announcement and S_t is the natural gas inventory surprise for day t . The inventory surprises are computed using either the Bloomberg consensus or the “informed” consensus. The “informed” consensus is the median value of ranked analyst forecasts. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table IV
Regression Results for Returns and Order Imbalances before Inventory Announcements

	Dependent variable: Return in [-90min, -5min] window	Dependent variable: Order Imbalance in [-90min, -5min] window
Intercept (a_0)	-0.17 (0.05)***	-137.73 (57.78)**
Slope (a_1)	-0.04 (0.01)***	-27.24 (7.46)***
R^2	0.09	0.04

The table reports estimates for the following model: $Y_t = a_0 + a_1 S_t + e_t$, where Y_t is the continuously compounded futures return (expressed in percent) or the order imbalance in the window from 90 minutes before to 5 minutes before the announcement and S_t is the natural gas inventory surprise for day t . The inventory surprises are computed using the Bloomberg consensus forecast. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table V
Predicting the Inventory Surprise

	No lags of surprise	$L = 1$	$L = 2$
Intercept (a_0)	0.64 (0.36)*	0.65 (0.36)*	0.65 (0.32)**
a_1	0.95 (0.12)***	0.96 (0.12)***	0.94 (0.10)***
$adj R^2$	0.22	0.22	0.22

The table reports estimates for the following regression model: $S_t = a_0 + a_1 P_t + \sum_{n=1}^L b_n S_{t-n} + e_t$. S_t is the natural gas inventory surprise for day t . The predictor, P_t , is the difference between the “informed” and the Bloomberg consensus forecasts. The “informed” consensus is the median value of ranked analyst forecasts. The sample period is from July 2008 to January 2016 and contains 334 observations. The regression is estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table VI
Relation between Predictor of the Inventory Surprise and Returns and Order Imbalances

	[-90min, -5min] window	[-5min, 30min] window
Panel A. Futures returns		
Intercept (a_0)	-0.20 (0.05)***	-0.31 (0.12)**
a_1	-0.06 (0.02)***	0.02 (0.04)
R^2	0.05	0.00
Panel B. Order imbalances		
Intercept (a_0)	-157.21 (57.74)***	-322.14 (132.3)**
a_1	-35.47 (13.99)**	30.11 (35.00)
R^2	0.02	0.00

Panel A reports estimates for the following model: $R_t = a_0 + a_1 P_t + e_t$, where R_t is the continuously compounded futures return (expressed in percent) in a given event window and P_t is the difference between the “informed” consensus and the Bloomberg consensus. The “informed” consensus is the median value of ranked analyst forecasts. Panel B reports estimates for a similar regression with the order imbalance used as the dependent variable. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table VII
Anchoring Bias in Inventory Forecasts

	$h = 1$	$h = 3$
Panel A: Surprise based on the consensus forecast		
γ	0.048 (0.014)***	0.004 (0.002)*
R^2	0.066	0.006
Panel B: Surprise based on the “informed” consensus forecast		
γ	0.014 (0.013)	0.001 (0.002)
R^2	-0.002	-0.008

The table reports estimates for the following model: $S_t = \gamma(F_t - \bar{A}_h) + \varepsilon_t$, where S_t is the inventory surprise, F_t is the consensus forecast, and $\bar{A}_h$ represents the historical average of the last h realizations of inventory changes. The “informed” consensus is the median value of ranked analyst forecasts. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table VIII
Regression of Pre-announcement Returns on Expected and
Unexpected Components of Inventory Surprises

	(1)	(2)
a_0	-0.17 (0.05)***	-0.19 (0.05)***
a_1	-0.04 (0.02)*	-0.04 (0.02)*
a_2	-0.04 (0.01)***	
R^2	0.09	0.01

Column (1) reports estimates for the following model: $R_t = a_0 + a_1 S_t^e + a_2 S_t^u + e_t$, where R_t is the continuously compounded futures return in the event window from 90 minutes before to 5 minutes before the announcement, and S_t^e and S_t^u are the anticipated and unanticipated components of natural gas inventory surprises. $S_t^e \equiv \gamma(F_t - \bar{A}_1)$ and $S_t^u \equiv S_t - \gamma(F_t - \bar{A}_1)$. S_t is the inventory surprise, F_t is the consensus forecast, and $\bar{A}_1$ is the previous realization of inventory change. $\gamma = 0.048$, as estimated in Table VII. Column (2) reports results for the regression with S_t^e only. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table IX
Comparison of the Anchoring Bias Predictor and the Predictor Based on Ranked Analyst Forecasts

	Dependent variable: Return in [-90min, -5min] window	Dependent variable: Inventory surprise
a_0	-0.20 (0.05)***	0.66 (0.36)*
a_1	0.00 (0.03)	0.02 (0.02)
a_2	-0.06 (0.02)***	0.86 (0.12)***
R^2	0.05	0.22

The table reports estimates for the following model: $Y_t = a_0 + a_1 S_t^e + a_2 P_t + e_t$, where Y_t is the continuously compounded futures return in the window from 90 minutes before to 5 minutes before the announcement or the inventory surprise based on the Bloomberg consensus, S_t^e is the anchoring bias predictor, and P_t is the predictor based on ranked analyst forecasts. $S_t^e \equiv \gamma(F_t - \bar{A}_1)$, where F_t is the Bloomberg consensus forecast and $\bar{A}_1$ is the previous realization of inventory change. $\gamma = 0.048$, as estimated in Table VII. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table X
Forecasting the Inventory Surprise and Pre-announcement Return with Predictor
Based on Shorter Evaluation Periods

	6 months	12 months	18 months
Panel A. Inventory Surprise			
Intercept (a_0)	0.38 (0.38)	0.47 (0.39)	0.42 (0.38)
Slope (a_1)	1.15 (0.24)***	1.16 (0.28)***	1.18 (0.25)***
R^2	0.09	0.08	0.09
Panel B. Return in [-90min, -5min] window			
Intercept (a_0)	-0.18 (0.05)***	-0.29 (0.05)***	-0.19 (0.05)***
Slope (a_1)	-0.07 (0.03)**	-0.06 (0.03)*	-0.07 (0.03)**
R^2	0.02	0.01	0.02

Panel A reports estimates for the following model: $S_t = a_0 + a_1 P_t^M + e_t$, where S_t is the natural gas inventory surprise based on Bloomberg consensus for day t . P_t^M is the difference between the median value of the top three economist forecasts and the Bloomberg consensus. Economists are ranked based on forecasting accuracy in the past 6, 12 and 18 months, respectively. Panel B reports estimates for the following model: $R_t = a_0 + a_1 P_t^M + e_t$, where R_t is the continuously compounded futures return (expressed in percent) from 90 minutes to 5 minutes before the announcement. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table XI
Forecasting the Inventory Surprise and Pre-announcement Return with Modified Predictor

	$M = 5$	$M = 10$
Panel A. Inventory Surprise		
Intercept (a_0)	0.53 (0.36)	0.48 (0.37)
Slope (a_1)	1.08 (0.14)***	1.16 (0.18)***
R^2	0.20	0.15
Panel B. Return in [-90min, -5min] window		
Intercept (a_0)	-0.19 (0.05)***	-0.19 (0.05)***
Slope (a_1)	-0.07 (0.02)***	-0.06 (0.02)***
R^2	0.05	0.03

Panel A reports estimates for the following model: $S_t = a_0 + a_1 P_t^M + e_t$, where S_t is the natural gas inventory surprise based on Bloomberg consensus for day t . P_t^M is the difference between the median value of the top M economist forecasts and the Bloomberg consensus. Panel B reports estimates for the following model: $R_t = a_0 + a_1 P_t^M + e_t$, where R_t is the continuously compounded futures return (expressed in percent) from 90 minutes to 5 minutes before the announcement. The sample period is from July 2008 to January 2016 and contains 334 observations. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

Table XII
Out-of-sample R^2 Statistics

	Dependent variable: Inventory Surprise	Dependent variable: Return in [-90min, -5min] window
Panel A. Mean Benchmark		
Out-of-sample period:		
January 2010-January 2016	14.42***	3.43***
January 2012-January 2016	13.51***	2.90**
January 2014-January 2016	21.77***	17.68***
Panel B. AR Benchmark		
AR(1)	15.31***	6.39***
AR(2)	14.56***	7.69***

Panel A reports the proportional reduction in mean square forecast error (R_{OS}^2) for predictive regression for the natural gas inventory surprise and the pre-announcement return based on the mean benchmark vis-à-vis competing forecast based on the predictor.

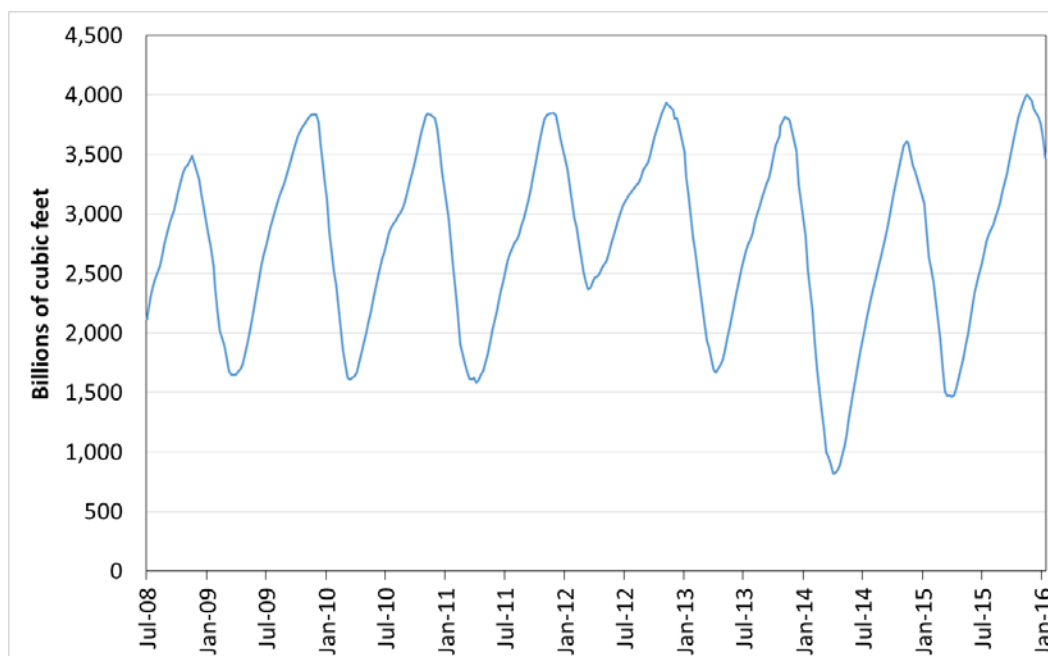
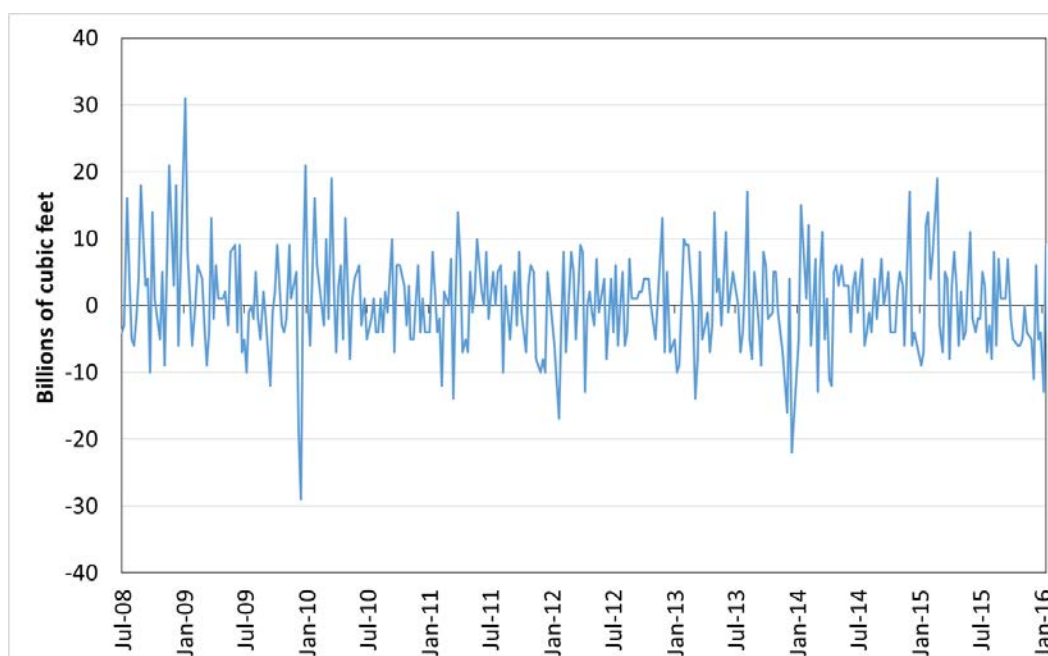
Panel B reports the proportional reduction in mean square forecast error (R_{OS}^2) for predictive regression for the natural gas inventory surprise and the pre-announcement return based on an AR model vis-à-vis the competing forecasting model including the predictor as an additional repressor. The out-of-sample period in Panel B is from January 2010 to January 2016.

Statistical significance is based on the Clark and West (2007) statistic for testing the null hypothesis that the MSFE of the baseline model is less than or equal to the competing model MSFE against the alternative hypothesis that the MSFE of the baseline model is greater than the competing model MSFE. The full sample period is from July 2008 to January 2016. *, **, *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

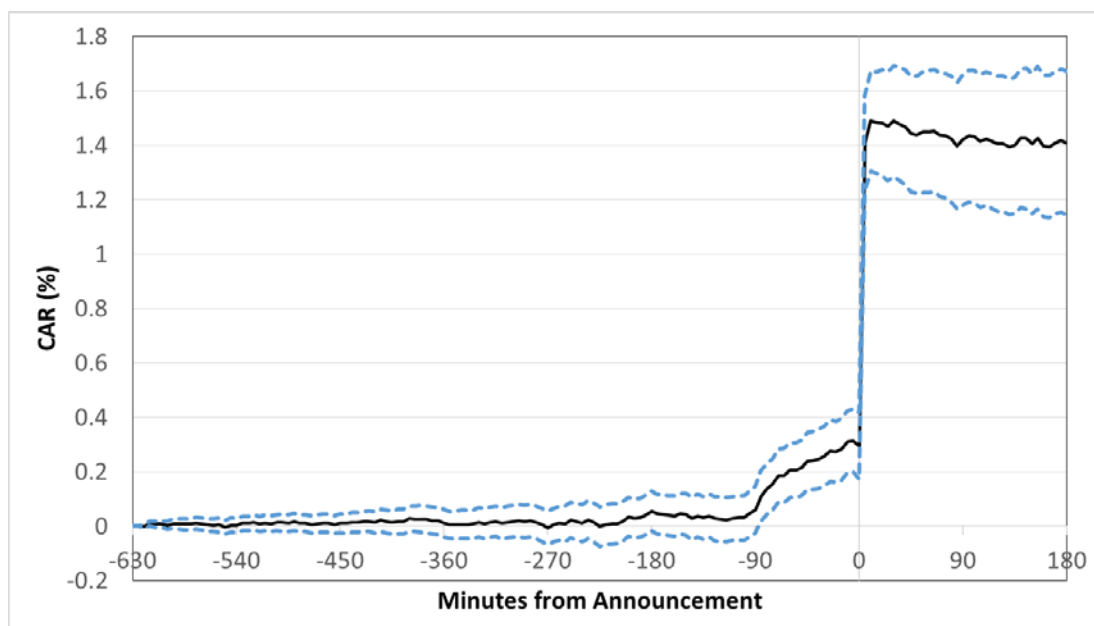
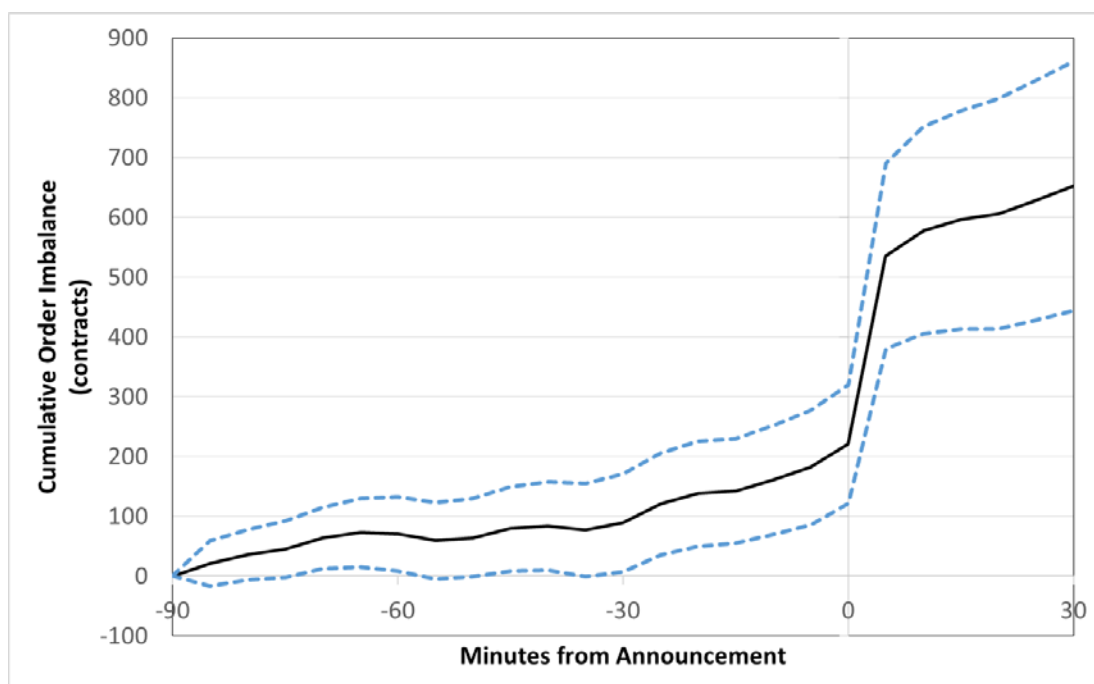
Table XIII
Performance of Trading Strategy Based on Predictor of the Inventory Surprise

	Full Sample (N=271)	Large Values of Predictor (N=153)
Sharpe Ratio	0.84	1.26
Sortino Ratio	1.40	2.12

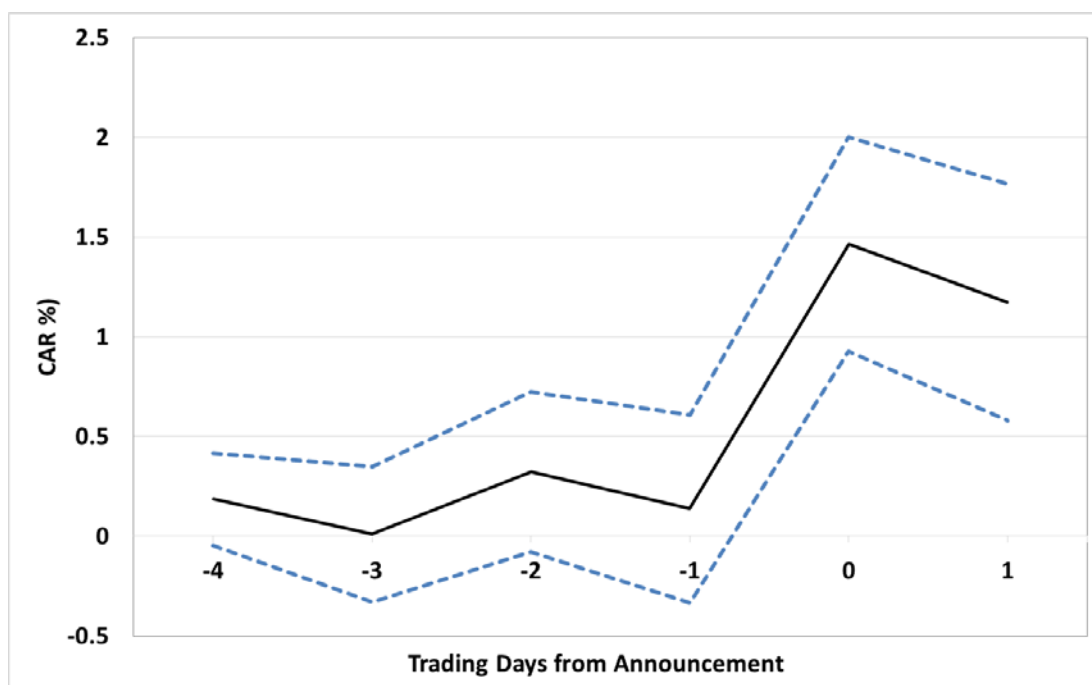
The table reports performance of the trading strategy that consists of buying (selling) a futures contract 90 minutes before the announcement if the predictor is negative (positive), and closing the position 5 minutes before the announcement. The predictor is defined as the difference between the “informed” consensus and the Bloomberg consensus forecast. The “informed” consensus is the median value of ranked analyst forecasts. The sample period is from July 2008 to January 2016 and contains 334 observations, including 271 announcements with non-zero values of the predictor. We classify non-zero values of the predictor in the 1st and 4th quartiles as large and the remaining values as small. We compute the annualized Sharpe ratio and Sortino ratio as the per-event ratio times the square root of the average number of events per year.

Figure 1. Natural Gas Inventory**Panel A: Natural Gas Inventory Level****Panel B: Natural Gas Inventory Surprises**

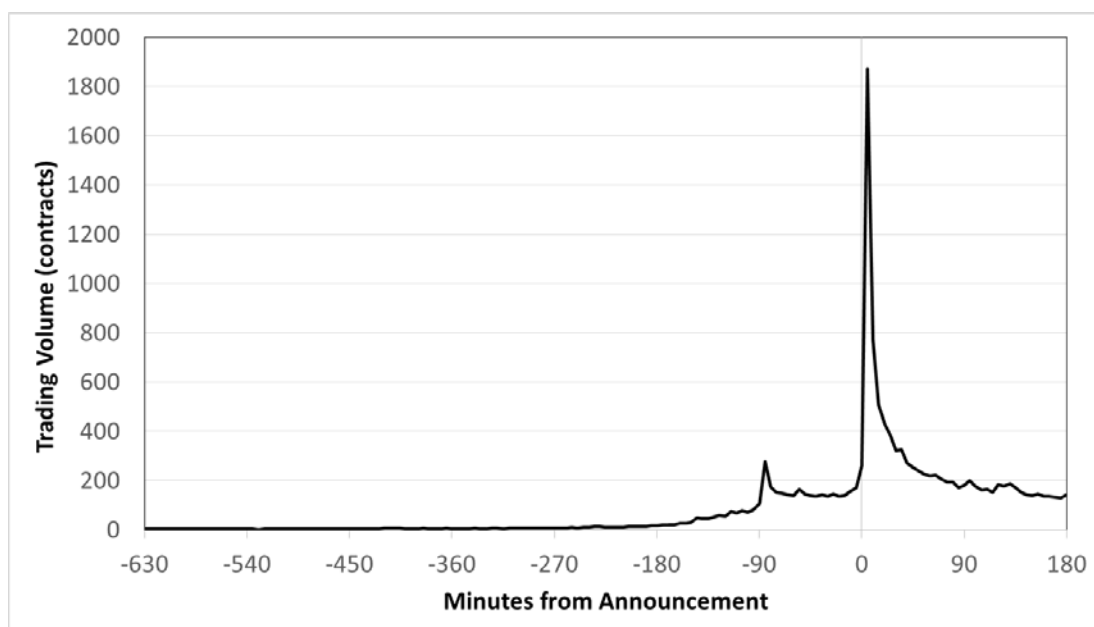
Panel A depicts the natural gas inventory level. Panel B shows the weekly natural gas storage surprises. The surprise is the difference between the announced change in inventory and the Bloomberg consensus forecast. The sample period is from July 2008 to January 2016 and contains 334 announcements.

Figure 2. Cumulative Average Returns and Order Imbalances**Panel A. Cumulative Average Returns****Panel B. Cumulative Average Order Imbalances**

Panel A shows the cumulative average returns of the nearby natural gas futures contract around the release of the Weekly Natural Gas Storage Report. Panel B shows the cumulative average order imbalances in the same futures contract. The dashed lines represent 90% confidence interval. The sample period is from July 2008 through January 2016 and contains 334 announcements.

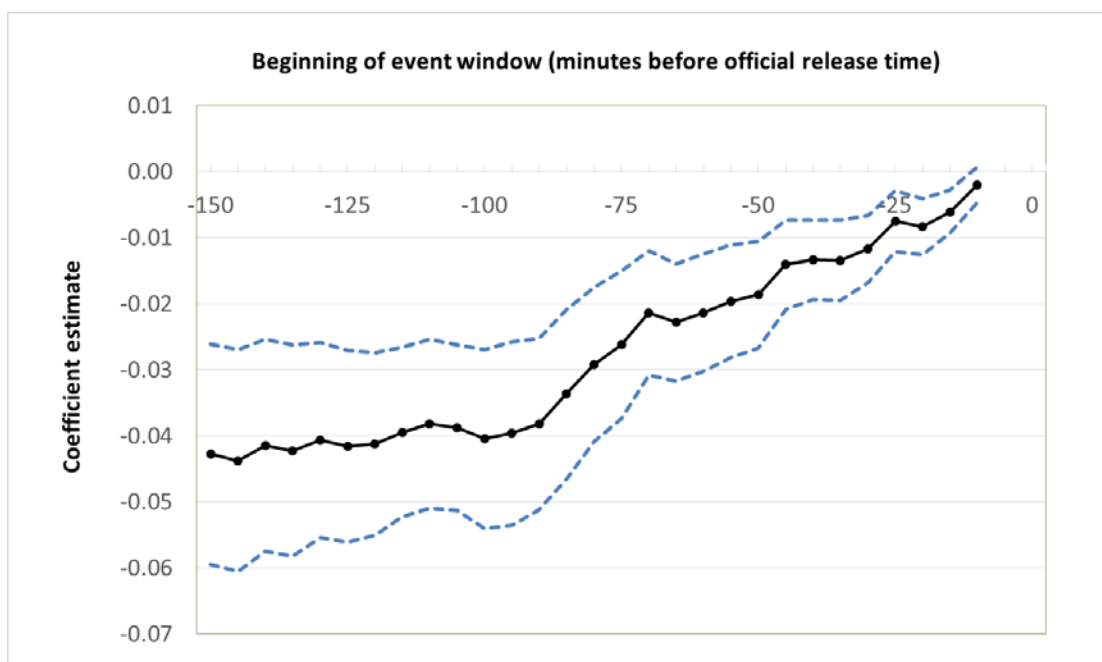
Figure 3. Daily Cumulative Average Returns

The figure shows the daily cumulative average returns of the nearby natural gas futures contract around releases of the Weekly Natural Gas Storage Report. Day 0 on the horizontal axis is the day of the announcement. The dashed lines represent 95% confidence interval. The sample period is from July 2008 through January 2016 and contains 334 announcements.

Figure 4. Average Trading Volume

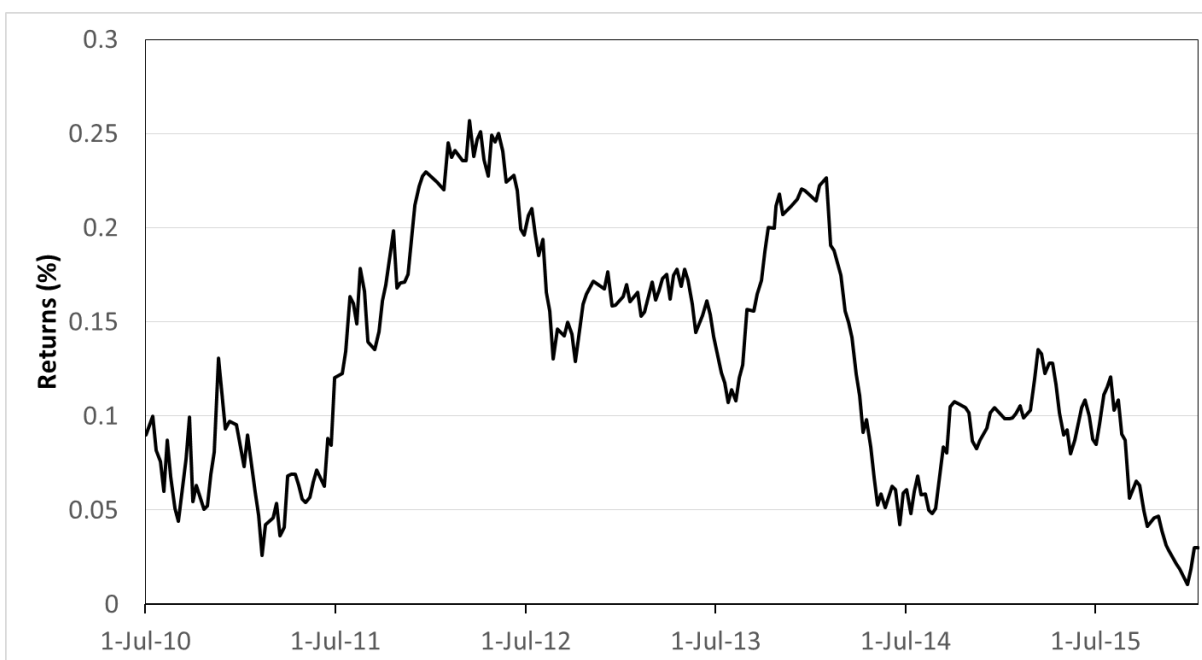
The figure plots the average trading volume per minute in the nearby natural gas futures contract around the release of the Weekly Natural Gas Storage Report. The sample period is from July 2008 through January 2016 and contains 334 announcements.

Figure 5. Sensitivity to Event Window Length



The figure plots the slope coefficient estimates from the OLS regression of natural gas futures returns on inventory surprises. The inventory surprise is computed using the Bloomberg consensus forecasts. The natural gas futures return is computed in the event window from the time shown on the horizontal axis to five minutes before the announcement. The dashed lines represent 95% confidence interval. The sample period is from July 2008 through January 2016 and contains 334 announcements.

Figure 6. Moving Average of Pre-announcement Trading Returns



The figure plots the 24-month moving average of the return of the trading strategy that consists of buying (selling) a futures contract 90 minutes before the announcement if the predictor is negative (positive), and closing the position 5 minutes before the announcement. The predictor is the difference between the “informed” consensus and the Bloomberg consensus forecast. The “informed” consensus is the median value of ranked analyst forecasts. The sample period is from July 2008 to January 2016 and contains 334 observations, including 271 announcements with non-zero values of the predictor.